

Revision Notes for Class 12 Physics

Chapter 2 – Electrostatic Potential and Capacitance

Electric Potential

Electrostatic potential at a point in space is the amount of work done in bringing a unit positive charge from infinity to that point in an electric field, without any acceleration. It is a scalar quantity and is denoted by V .

The formula for electrostatic potential V due to a point charge q at a distance r from the charge is given by:

$$V = \frac{kq}{r} \text{ where:}$$

- V = Electrostatic potential,
- k = Coulomb's constant ($9 \times 10^9 \text{ Nm}^2 / \text{C}^2$),
- q = Source charge,
- r = Distance from the source charge.

The SI unit of electrostatic potential is the volt (V). One volt is equivalent to one joule per coulomb ($1 \text{ V} = 1 \text{ J/C}$), which means that 1 volt represents the potential difference when 1 joule of work is done to move 1 coulomb of charge between two points in an electric field.

Electrostatic Potential Difference Electrostatic Potential Difference between two points in an electric field is the amount of work done in moving a unit positive charge from one point to another. It represents the difference in electrostatic potential between the two points and is a scalar quantity.

Formula:

The potential difference ΔV between two points A and B in an electric field is given by:

$$\Delta V = V_B - V_A = \frac{W_{AB}}{q}$$

where:

- ΔV = Potential difference between points A and B ,
- V_B and V_A = Electrostatic potentials at points B and A , respectively,
- W_{AB} = Work done in moving a charge q from A to B ,
- q = Magnitude of the charge.

Properties of Electrostatic Potential Difference:

1. **Scalar Quantity:** The potential difference is a scalar quantity and does not have direction, only magnitude.
2. **Independent of Path:** In an electrostatic field, the potential difference between two points depends only on the position of the points and not on the path taken by the charge.
3. **Work Done:** It measures the work required to move a unit positive charge between two points. A positive potential difference implies work is done against the field, while a negative potential difference implies the field does work.
4. **SI Unit:** The SI unit of electrostatic potential difference is the volt (V).
5. **Relation to Electric Field:** The potential difference between two points is related to the electric field by the equation:

$\Delta V = -\int_A^B \vec{E} \cdot d\vec{r}$ where $\vec{E}$ is the electric field and $d\vec{r}$ is the displacement vector along the path from A to B.

Electric Potential due to a Point Charge, a Dipole, and a System of Charges :

1. Electric Potential Due to a Point Charge:

- The electric potential at a distance r from a point charge q is the amount of work done to bring a unit positive charge from infinity to the point.

- Formula:

$$V = \frac{kq}{r}$$

where:

- V = Electric potential,

- k = Coulomb's constant ($9 \times 10^9 \text{ Nm}^2 / \text{C}^2$),

- q = Charge of the point,

- r = Distance from the charge.

2. Electric Potential Due to a Dipole:

- A dipole consists of two equal and opposite charges separated by a small distance.

- The potential at a point due to an electric dipole depends on the position relative to the dipole axis.

- Formula:

$$V = \frac{1}{4\pi\epsilon_0} \cdot \frac{p \cos \theta}{r^2}$$

where:

- V = Electric potential,
- p = Dipole moment ($p = q \times d$, charge q and separation d),
- θ = Angle between the position vector and the dipole axis,
- r = Distance from the center of the dipole,
- ϵ_0 = Permittivity of free space.

3. Electric Potential Due to a System of Charges:

- For a system of charges , the total electric potential at a point is the algebraic sum of the potentials due to individual charges.

- Formula:

$$V = \frac{1}{4\pi\epsilon_0} \sum_{i=1}^n \frac{q_i}{r_i}$$

where:

- V = Total electric potential at a point,
- q_i = Individual charges in the system,
- r_i = Distance of each charge q_i from the point,

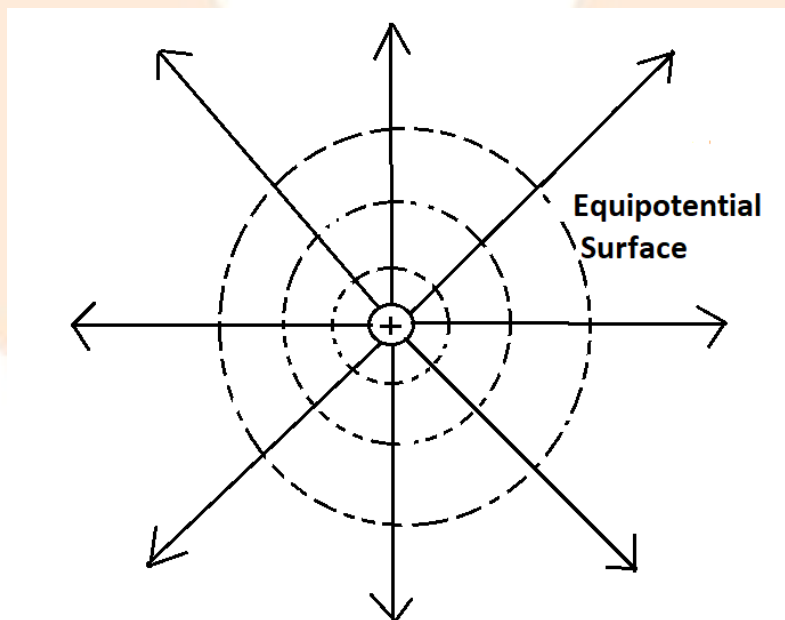
- ϵ_0 = Permittivity of free space.

Key Points:

- Point Charge: Potential varies inversely with distance r .
- Dipole: Potential depends on both distance r and angle θ .
- System of Charges: The potential is the sum of potential due to individual charges.

Equipotential Surface

An equipotential surface is a surface where the electric potential at every point is the same. This means that if a charge is moved along this surface, no work is done by or against the electric field because the potential difference between any two points on the surface is zero.



Key Properties of Equipotential Surfaces:

1. No Work Done: Moving a charge along an equipotential surface requires no work, as the potential difference is zero.
2. Perpendicular to Electric Field: Equipotential surfaces are always perpendicular to the electric field lines. This is because work would otherwise be done if the charge moved in the direction of the field.
3. Closer Surfaces, Stronger Field: In regions where the equipotential surfaces are closer together, the electric field is stronger. This is because the electric potential changes more rapidly over a small distance.
4. Shape Depends on Charge Distribution:
 - For a point charge, the equipotential surfaces are concentric spheres around the charge.
 - For a uniform electric field, the equipotential surfaces are parallel planes.
 - For a dipole, the surfaces are more complex, but still symmetric.

Relation between field and potential

The relationship between electric field E and electric potential V is an important concept in electrostatics. The electric field is the negative gradient of the electric potential. In simpler terms, the electric field represents how quickly and in what direction the potential is changing in space.

Formula:

$$E = -\frac{dV}{dr} \text{ where:}$$

- E = Electric field strength,

- dV = Change in electric potential,

- dr = Small displacement in the direction of the field.

Potential Energy in an External Electric Field refers to the energy stored in a charge due to its position in an external electric field. This energy arises because of the interaction between the charge and the external electric field. The potential energy of a charge depends on its location within the field.

1. Potential Energy of a Point Charge in an External Electric Field:

The potential energy U of a point charge q placed at a point in an external electric field E is given by:

$$U = qV$$

For a uniform electric field, where $V = Ed$ (with E being the electric field and d being the displacement from a reference point), the potential energy becomes:

$$U = qEd$$

2. Potential Energy of an Electric Dipole in an External Electric Field:

An electric dipole in an external uniform electric field also has potential energy due to the interaction between the field and the dipole moment p . The potential energy of the dipole is given by:

$$U = -pE \cos \theta$$

Key Points:

- A positive charge in an electric field moves from a region of higher potential to lower potential, and the potential energy decreases as it moves in the direction of the field.

- A negative charge moves from a region of lower potential to higher potential, and its potential energy increases as it moves against the direction of the field.
- For a dipole, the potential energy is minimum when the dipole is aligned with the electric field ($\theta = 0^\circ$) and maximum when the dipole is perpendicular to the field ($\theta = 90^\circ$).

Conductors

Conductors are materials that allow the free flow of electric charges (usually electrons) through them. In a conductor, the outer electrons of atoms are loosely bound and can move easily within the material when an electric field is applied. Conductors are used extensively in electrical circuits and devices because of their ability to carry electric current efficiently.

Key Properties of Conductors:

1. **Free Electrons:** Conductors have free electrons (also called conduction electrons) that are not tightly bound to any specific atom, allowing them to move freely throughout the material.
2. **Electric Field Inside a Conductor:** In the electrostatic condition (when the charges are not moving), the electric field inside a conductor is **zero**. This is because free electrons move in such a way as to cancel any applied electric field within the conductor.
3. **Charge Distribution:** In a conductor, excess charges (if any) reside on the surface of the conductor. This occurs because like charges repel each other and spread out as far as possible, ending up on the outer surface of the conductor.
4. **Equipotential Surface:** The entire conductor acts as an equipotential surface in electrostatic equilibrium, meaning that the potential at every point within and on the surface of the conductor is the same.
5. **Electric Field at the Surface:** The electric field just outside the surface of a conductor is perpendicular to the surface. If the surface is irregular, the electric field is stronger at points where the surface curvature is high (such as sharp edges or points).

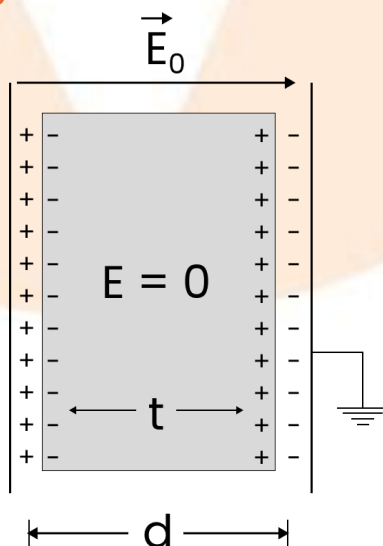
6. **Electrostatic shielding** refers to the phenomenon where a conductor blocks or shields the interior from external electric fields. This occurs because the electric field inside a conductor is zero in electrostatic equilibrium, meaning that any charges or electric fields outside the conductor cannot affect the inside.

Conductors in Electric Fields:

- When a conductor is placed in an external electric field, free electrons move and rearrange themselves to cancel the field inside the conductor. This creates an induced charge on the surface of the conductor, resulting in an electric field outside the conductor.

Dielectrics and Polarisation

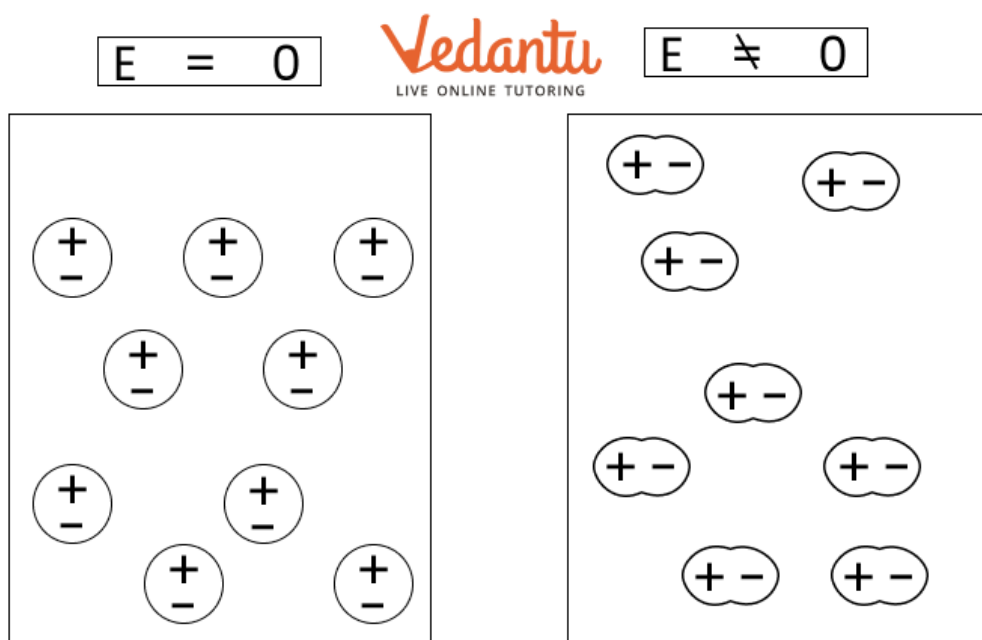
Dielectrics are insulating materials that do not conduct electricity but can be polarized when placed in an external electric field. In simple terms, a dielectric is a substance in which an electric field can induce dipole moments, but it does not allow the free movement of charge like conductors do.



Polarisation in Dielectrics:

When a dielectric material is placed in an external electric field, its molecules become polarized, meaning that their positive and negative charges get slightly separated. This phenomenon is called polarisation.

- Polarisation (P) is the alignment of electric dipoles in the material due to the external field. It is defined as the dipole moment per unit volume of the dielectric.



Types of Polarisations:

1. **Electronic Polarisation:** This occurs when the electric field displaces the electron cloud relative to the nucleus in atoms. It is common in all dielectrics.
2. **Ionic Polarisation:** In materials with ionic bonds, the electric field causes the positive and negative ions to shift slightly in opposite directions.
3. **Orientational (Dipolar) Polarisation:** In dielectrics with permanent dipole moments (like water), the external electric field aligns the dipoles along the direction of the field.

Polarisation Vector:

The Polarisation $\mathbf{P}$ is a vector quantity representing the dipole moment per unit volume:

$$\mathbf{P} = \frac{\text{Total Dipole Moment}}{\text{Volume}}$$

Effect of Polarisation:

1. **Reduced Electric Field:** The electric field inside a dielectric is weaker than in a vacuum because the induced dipoles produce an internal electric field that opposes the external field. This reduces the overall field strength within the dielectric.
2. **Dielectric Constant (Relative Permittivity) :** The reduction in the electric field due to polarisation is quantified by the dielectric constant ϵ_r , which is the ratio of the electric field in a vacuum to that in the dielectric:

$$\epsilon_r = \frac{E_0}{E}$$

where E_0 is the external electric field and E is the reduced field within the dielectric.

3. **Capacitance Increase:** When a dielectric is placed between the plates of a capacitor, it increases the capacitor's ability to store charge by reducing the electric field and increasing capacitance:

$$C = \epsilon_r \cdot C_0$$

where C_0 is the capacitance without the dielectric.

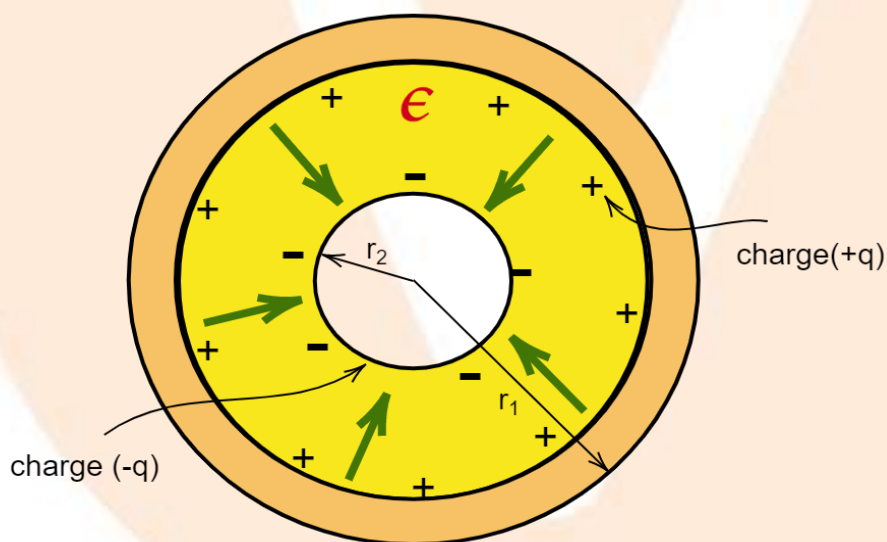
Key Properties of Dielectrics:

- **Insulating Material:** Dielectrics do not conduct electricity but can become polarized.

- Dielectric Strength: This is the maximum electric field a dielectric can withstand without breaking down (allowing current to pass through).
- Induced Charge: The charges induced by Polarisation are called bound charges and appear on the surface of the dielectric.

Capacitors and Capacitance

Capacitors are devices that store electrical energy in an electric field. They consist of two conductive plates separated by an insulating material called a dielectric. When a voltage is applied across the plates, an electric field is created, and charge accumulates on the plates, leading to energy storage.



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Capacitance:

Capacitance is the ability of a capacitor to store charge per unit voltage. It is a measure of how much electric charge a capacitor can hold at a given potential difference. Capacitance depends on the size of the plates, the distance between them, and the properties of the dielectric material.

The formula for capacitance is:

$$C = \frac{Q}{V}$$

Capacitance of a Parallel Plate Capacitor:

for a parallel plate capacitor, the capacitance is given by:

$$C = \epsilon_0 \epsilon_r \frac{A}{d}$$

where:

- C = Capacitance,
- ϵ_0 = Permittivity of free space (8.85×10^{-12} F/m),
- ϵ_r = Relative permittivity (dielectric constant) of the material between the plates,
- A = Area of one of the plates,
- d = Distance between the plates.

Key Points About Capacitors:

1. Energy Stored in a Capacitor: The energy U stored in a capacitor is given by:

$$U = \frac{1}{2} CV^2$$

This represents the energy stored in the electric field between the plates.

2. Factors Affecting Capacitance:

- Plate Area (A): Increasing the area of the plates increases the capacitance.
- Distance Between Plates (d): Decreasing the distance between the plates increases the capacitance.
- Dielectric Material: Introducing a dielectric material with a higher dielectric constant (ϵ_r) increases the capacitance.

Types of Capacitors:

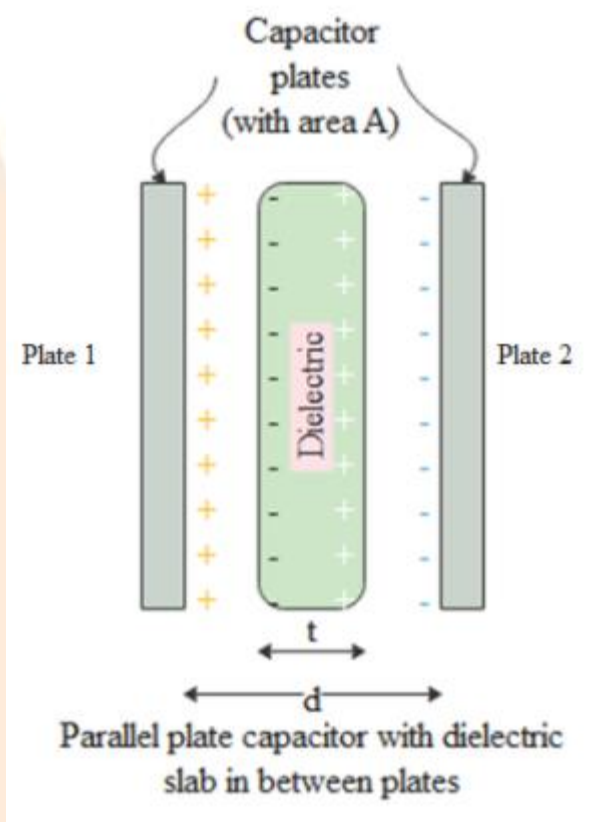
- Ceramic Capacitors: Small, non-polarized capacitors made from ceramic materials.
- Electrolytic Capacitors: Polarized capacitors with high capacitance values, used in applications like power supplies.
- Tantalum Capacitors: Similar to electrolytic capacitors but offer higher stability and reliability.

Applications of Capacitors:

1. Energy Storage: Capacitors store electrical energy for use in circuits like flashlights or camera flashes.
2. Filtering: Capacitors filter out noise in power supplies and signal processing.
3. Timing Circuits: Capacitors are used in combination with resistors to create time delays.
4. Coupling and Decoupling: Capacitors allow AC signals to pass through while blocking DC signals in circuits.

Parallel Plate Capacitor

A parallel plate capacitor consists of two large, flat conductive plates placed parallel to each other and separated by a small distance. When a potential difference (voltage) is applied across the plates, one plate becomes positively charged, and the other becomes negatively charged. This creates an electric field between the plates, allowing the capacitor to store energy in the form of electric potential energy.



Capacitance of a Parallel Plate Capacitor:

The capacitance C of a parallel plate capacitor depends on three factors:

1. Area of the plates A ,
2. Distance between the plates d ,

3. Dielectric material between the plates (with relative permittivity ϵ_r).

The formula for the capacitance of a parallel plate capacitor is:

$$C = \epsilon_0 \epsilon_r \frac{A}{d} \text{ where:}$$

- C = Capacitance (in farads, F),

- ϵ_0 = Permittivity of free space (8.85×10^{-12} F/m),

- ϵ_r = Relative permittivity (dielectric constant) of the dielectric material,

- A = Area of one of the plates (in square meters),

- d = Distance between the plates (in meters).

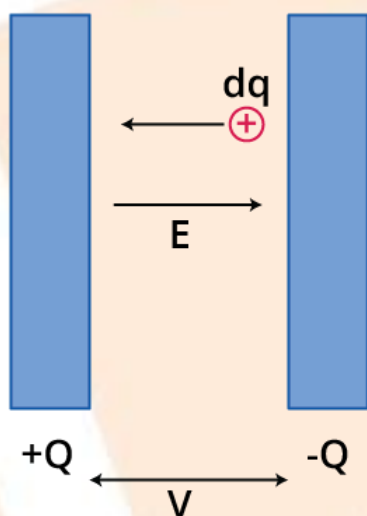
Key Concepts:

1. Electric Field Between Plates:

The electric field E between the plates of a parallel plate capacitor is uniform and is given by:

$$E = \frac{V}{d}$$

where V is the potential difference across the plates, and d is the distance between them.



2. Charge Stored:

The charge Q stored on the plates of the capacitor is related to the capacitance and voltage by:

$$Q = C \cdot V$$

Here, Q is the charge on the plates, C is the capacitance, and V is the voltage applied across the plates.

3. Energy Stored:

The energy U stored in a parallel plate capacitor is the electric potential energy stored in the electric field between the plates. The energy is given by:

$$U = \frac{1}{2} CV^2$$

Effect of a Dielectric:

When a dielectric material (an insulating material) is placed between the plates of a capacitor, it increases the capacitance. The dielectric reduces the electric field within the capacitor, allowing more charge to be stored for the same applied voltage.

- The dielectric constant ϵ_r increases the capacitance by a factor of ϵ_r . For a vacuum, $\epsilon_r = 1$, while for most materials, $\epsilon_r > 1$.

Key Factors Affecting Capacitance:

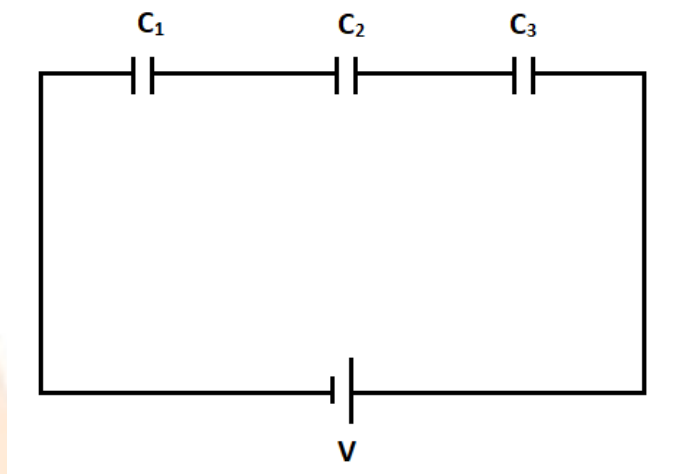
1. Plate Area (A): Increasing the area of the plates increases the capacitance because more charge can be stored.
2. Distance Between Plates (d): Decreasing the distance between the plates increases the capacitance because the electric field becomes stronger.
3. Dielectric Material: Inserting a dielectric with a higher dielectric constant (ϵ_r) increases the capacitance.

Combination of Capacitors

Capacitors can be connected in different ways in a circuit to achieve desired capacitance values. The two most common configurations are series combination and parallel combination. Each combination affects the total or equivalent capacitance of the system differently.

1. Capacitors in Series:

When capacitors are connected in series, the total capacitance is reduced, and the equivalent capacitance is always less than the smallest capacitance in the series.



- Formula for Series Combination:

$$\frac{1}{C_{\text{total}}} = \frac{1}{C_1} + \frac{1}{C_2} + \dots + \frac{1}{C_n}$$

where $C_1, C_2, \dots, C_n$ are the individual capacitances of the capacitors connected in series, and C_{total} is the equivalent capacitance.

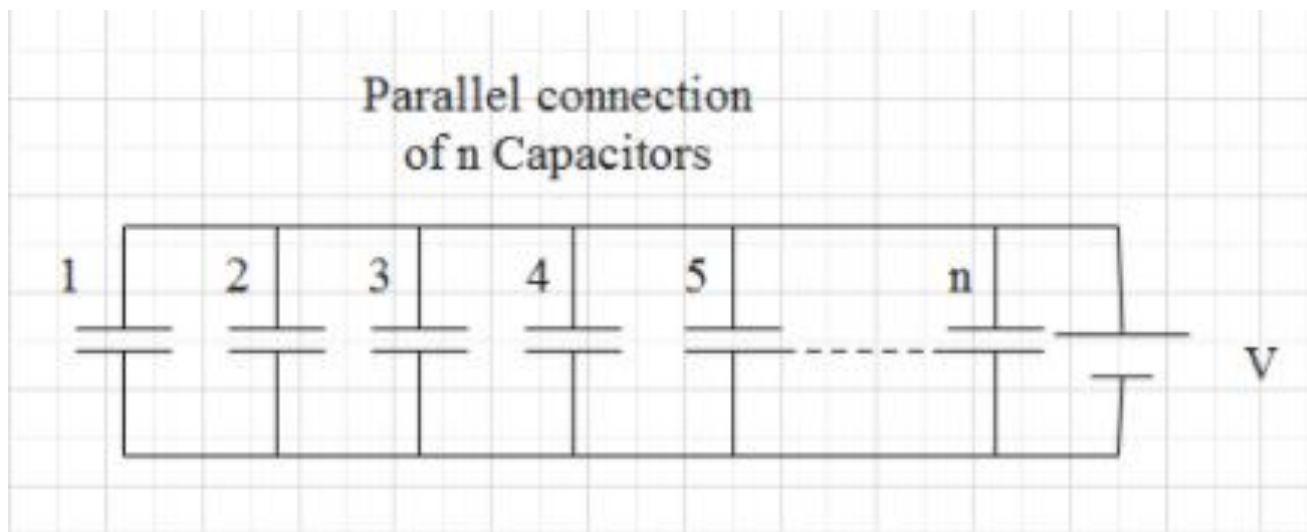
- Charge: In a series circuit, the charge Q on each capacitor is the same because the same current flows through all the capacitors.

- Voltage: The total voltage across the series combination is the sum of the individual voltages across each capacitor:

$$V_{\text{total}} = V_1 + V_2 + \dots + V_n$$

2. Capacitors in Parallel:

When capacitors are connected in parallel, the total capacitance increases, and the equivalent capacitance is the sum of the individual capacitances. The total capacitance is greater than the largest individual capacitance.



- Formula for Parallel Combination:

$$C_{\text{total}} = C_1 + C_2 + \dots + C_n$$

where $C_1, C_2, \dots, C_n$ are the individual capacitances, and C_{total} is the equivalent capacitance.

- Voltage: In a parallel circuit, the voltage across each capacitor is the same:

$$V_1 = V_2 = \dots = V_n$$

- Charge: The total charge stored by the combination is the sum of the charges on each capacitor:

$$Q_{\text{total}} = Q_1 + Q_2 + \dots + Q_n$$

Key Differences Between Series and Parallel Combinations:

Property	Series Combination	Parallel Combination
Capacitance	Decreases: $\frac{1}{C_{\text{total}}} = \sum \frac{1}{C_i}$	Increases: $C_{\text{total}} = \sum C_i$
Charge	Same charge on each capacitor: $Q_1 = Q_2 = Q_n$	Charge is divided across capacitors: $Q_{\text{total}} = \sum Q_i$
Voltage	Sum of individual voltages: $V_{\text{total}} = V_1 + V_2$	Same voltage across each capacitor: $V_1 = V_2 = \dots = V_n$
Application	Used to increase voltage handling capability	Used to increase total charge storage

3. Mixed (Series-Parallel) Combination:

In more complex circuits, capacitors may be connected in both series and parallel combinations. To find the equivalent capacitance in such circuits:

1. Reduce the parallel groups into a single capacitance using the parallel formula.
2. Reduce the resulting series groups using the series formula.
3. Repeat the process until the entire circuit is simplified to a single equivalent capacitance.

Applications:

- Series Combination: Used where higher voltage handling is needed, like in high-voltage circuits, as the total voltage is distributed across the capacitors.
- Parallel Combination: Used to increase the total capacitance in circuits where more charge storage is required, such as in power supply filters.

Energy Stored in a Capacitor

A capacitor stores energy in the form of an electric field between its plates when it is charged. The energy stored in a capacitor is related to the charge on the plates and the voltage across the plates. This energy can be used later when the capacitor is discharged.

Formula for Energy Stored:

The energy U stored in a capacitor is given by:

$$U = \frac{1}{2} CV^2 \text{ where:}$$

- U = Energy stored (in joules, J),
- C = Capacitance of the capacitor (in farads, F),
- V = Voltage across the capacitor (in volts, V).

Alternative Forms of the Formula:

Depending on the known quantities (charge, capacitance, or voltage), the energy stored in a capacitor can also be expressed in other forms:

1. Using charge Q and capacitance C :

$$U = \frac{Q^2}{2C}$$

2. Using charge Q and voltage V :

$$U = \frac{1}{2} QV$$

Explanation:

- The energy stored in a capacitor is proportional to the square of the voltage applied and directly proportional to the capacitance.
- This energy is stored in the electric field created between the plates of the capacitor when it is charged.
- When the capacitor is connected to a circuit, this stored energy can be released to power components.

Key Points:

1. Energy Density: The energy per unit volume stored in the electric field between the plates of a capacitor is called energy density. It is given by:

$$u = \frac{1}{2} \epsilon E^2$$

where ϵ is the permittivity of the dielectric material and E is the electric field strength.

2. Energy in a Discharging Capacitor: When a capacitor discharges, the energy stored in it is released, and the voltage across the capacitor decreases over time.

3. Applications: Capacitors are used in applications like power supplies, camera flashes, and backup power systems where energy storage and quick discharge are required.

In a camera flash, a capacitor stores electrical energy while it charges. When the flash is triggered, the stored energy is quickly released, producing a bright flash of light.